Response to Reviewers — Submission 6222

Original title: From Colonial Spatial Structures to Sovereign Spatial Planning: Multimodal Network Modelling and Optimal Location of New Urban Hubs in the Democratic Republic of the Congo Using Spatial Data Science.

Revised title: Urban Hub Coverage and Road Accessibility in Western Democratic Republic of the Congo: A Geospatial Analysis of Data and Model Sensitivity

Author: PAMBOLO MAYAMBO Néhémie. Revision extended: 7 October 2026.

Dear Editor and Reviewers, thank you for identifying substantive gaps between the original framing and the implemented analysis. The manuscript has been rewritten as an exploratory analysis of existing-hub road coverage and model sensitivity. We have removed historical causal claims, unsupported travel-time estimates, and new-town recommendations; added complete gridded demand, an alternative population product, a directed mapped-road graph, exact optimization, and joint stress tests; and clarified the candidate-score and hydrological audits. We do not claim that all requests have been fulfilled. In particular, a compatible regional empirical validation dataset remains unavailable. The new extension also compares 10/25-km greenfield screening grids and extreme tree rules with the October 2026 protected-area view, but does not establish forest integrity or constructability. We respectfully ask that the revision be assessed in light of this explicitly reduced scope.

1. Title, introduction, and the colonial-legacy claim

Reviewer comment: The reviewer requested either removal of the colonial claim from the title or a quantitative historical comparison.

Response: Addressed by removal and scope correction. The title now identifies existing-hub road coverage and sensitivity. Section I states that no colonial-era network is digitized, and Section III.F explains why current coverage cannot apportion fragmentation between colonial inheritance and post-independence investment. Historical motivation is no longer a model result. No invented diachronic comparison is supplied.

Location and evidence: Sections I, III.F, IV; revised title.

2. MST + three-neighbour topology and centrality

Reviewer comment: The reviewer requested sensitivity to alternative graph topologies and distance thresholds, including Kenge and Bandundu.

Response: New calculations compare MST + k for k = 1, 2, 3, 4, 5, and 8 and radius graphs at 15, 25, 40, 50, and 100 km. Kenge changes from betweenness rank 1 to 7 and Bandundu from 2 to 14 between k = 3 and 5. Radius 100 km yields 12 components and eight isolates. The centrality-based investment interpretation is withdrawn. The main model now uses mapped road geometry and preserves disconnection rather than forcing MST links. Closeness and reachability are exported for every node. Missing junction and restriction information remains a graph-quality limitation.

Location and evidence: Sections II.C, II.F, III.B, III.D; Fig. 3; Table VII; topology_node_metrics.csv and topology_summary.csv.

3. Sovereign hubs and functional polycentric planning

Reviewer comment: The reviewer asked how selected nodes can act as sovereign hubs and break dependency on Kinshasa.

Response: The empirical model does not measure economic autonomy, employment, productive specialization, energy reliability, or trade dependence. These claims have therefore been removed from the title and results. Section III.E explains why Yanganga and Ebale are not validated sovereign hubs, while III.F distinguishes selecting several facilities from functional polycentricity. The main revised objective concerns existing candidates and population coverage. No added sovereignty score is presented as measured evidence.

Location and evidence: Sections I, II.D, III.E–F, IV.

4. Hydrological proxy specification

Reviewer comment: The reviewer requested the formula or GIS procedure behind high/moderate exposure.

Response: The original cached rule is recovered and specified: high if distance to mapped water ≤5 km and slope ≤2%; otherwise moderate if distance ≤15 km and slope ≤5%; otherwise low, with missing inputs unclassified. The five-query SRTM90 kilometre-scale slope calculation is documented. We explicitly distinguish this heuristic from flood probability, depth, discharge, or a return-period map. The flag remains in the legacy audit and is not used to certify construction safety.

Location and evidence: Section II.F; candidate_weight_audit_metadata.json; original environment script retained in the released package.

5. Empirical validation using secondary observations

Reviewer comment: The reviewer requested validation against actual road, port, flight, field, or historical data.

Response: Not fully addressed. The JICA Kinshasa April 2019 master-plan report and appendices were downloaded; its survey-route inventory and Table 2.3.18 observed screen-line counts were examined. Two genuine observations (311,333 and 335,747 PCU/day) are retained. Our model does not predict PCU/day; no matching validation target is available. JICA's own estimated counts are not attributed to this study. It contains urban route observations, but these do not provide a compatible ten-province OD matrix or target for the regional 100-km coverage objective. No regional predictive error or field validation is reported. We removed the previous phrase "indirect validation": source consistency, recovery of names, and robustness are now diagnostics and computational verification. This remains a substantive external-validity limitation; the author does not ask the editor to treat the calculations as observed mobility or validated operational planning.

Location and evidence: Sections II.G, III.F, IV; reference [32]; jica_survey_route_inventory.csv. A matched empirical validation test is still required for stronger claims.

6. Joint uncertainty rather than weights alone

Reviewer comment: The reviewer requested simultaneous uncertainty in population, OSM completeness, speeds, and coordinates.

Response: Partially addressed by new joint calculations: two population products, one- and five-kilometre demand grids, main roads versus inclusion of tracks, connector limits 1/2.5/5 km, fixed 1% road-feature removal, and fixed 250-m hub-coordinate displacement. The full design has 96 configurations per p and 288 exact runs. For p = 10, coverage spans 40.40–63.35%. This envelope is not a confidence interval, and removal/displacement levels are uncalibrated stress assumptions. We withdrew travel-time claims, so no assumed speeds enter the revised objective. Road additions and distributions of positional or operational error are not calibrated. The extension adds 108 exact runs: an unperturbed baseline and five independent combined deletion/displacement realizations, both products, three hub counts, and 50/100/150-km coverage thresholds, at 5-km demand resolution. These tests remain uncalibrated scenario stresses, not confidence intervals or Sobol attribution.

Location and evidence: Section II.E; Tables II and VI; joint_maximal_coverage_results.csv and joint_coverage_metadata.json.

7. Justification of multicriteria weights

Reviewer comment: The reviewer requested empirical justification or expert elicitation for the criterion weights.

Response: The manuscript now labels the weights as assumptions, not expert elicitation or AHP. The audit distinguishes the six preliminary weights from the actual seven-variable environmental Monte Carlo vector. Repeating 5,000 draws around uniform weights changes dominant winners in three provinces. Crucially, Yanganga is the only compatible cached candidate in Tshuapa, making its 100% selection frequency automatic. These findings remove the previous robustness endorsement. No unperformed consultation is claimed; greenfield recommendations are withdrawn.

Location and evidence: Sections II.F and III.E; Table VIII; candidate_reference_uniform_weights.csv.

8. Population alternatives, road completeness, thresholds, grid resolution, forest definition, and Sobol analysis

Reviewer comment: The reviewer specified WorldPop versus alternatives, segment perturbations, thresholds 15/25/40/100 km, grid resolution, forest definition, and possible Sobol decomposition.

Response: The population alternatives and radius topology tests are completed as previously described. The extension now also compares complete 10- and 25-km candidate screens on a shared boundary mask with a common zero origin and recomputed complete-grid population potential. Native WorldCover samples compare strict tree exclusion with an extreme tree-admitted diagnostic. The official WDPCA protected-area view for October 2026 is applied with centre and full-cell intersection rules. The two grids yield 6,519 and 1,047 access/remoteness-eligible centres, with 1,298 and 268 full-cell protected-polygon intersections before land-cover screening. These calculations address the previously missing grid and hard-forest-rule sensitivity. They do not distinguish primary from degraded forest, certify tenure or flood safety, or constitute a Sobol decomposition. The strict full-cell screen retains 1,583 ten-kilometre candidates and 241 twenty-five-kilometre candidates. Équateur has no retained candidate on the latter grid. Among nine common provinces, the median top-centre shift is 212.22 km under reference weights and 52.02 km under uniform weights, demonstrating substantial grid sensitivity. Road additions and calibrated uncertainty distributions remain untested.

Location and evidence: Sections II.B, II.E–F, III.E–F; Table II; full supplementary run tables. Grid and hard-tree-rule comparisons are now computed; forest-integrity and field validation remain unperformed. Extension: Sections II.H and III.G, Tables IX–XI.

9. Complete population demand, standard urban definitions, and falsifiable elbow

Reviewer comment: The reviewer requested population cells or the full locality set, standard urban definitions, and a falsifiable hypothesis.

Response: The principal model now includes every positive selected population cell: 219,279 WorldPop one-kilometre cells (24,553 five-kilometre cells) and 147,956 GHS-POP one-kilometre cells (21,120 five-kilometre cells). Rural and disconnected demand remain in the denominator. GHS-SMOD point classifications replace fixed-radius interpretations; they do not estimate full agglomeration extents. The old p-median is retained only as an audit, solved exactly for p = 2–20. A specified three-successive-relative-gains-below-5% rule identifies no elbow; the earlier ten-hub claim is withdrawn. The regional model changes to maximal covering because the mapped-road graph is disconnected.

Location and evidence: Sections II.B, II.D, II.F, III.A, III.C–D; Tables III, V; Figs. 2, 4; elbow_rule_check.json.

Reproducibility and release status

The accompanying revision package contains the new Python scripts, verified derived tables, metadata, and figures. The existing Zenodo v1.0.0 archive (10.5281/zenodo.22287394) predates the revision and must not be represented as containing these new analyses. The original archive is retained unchanged. A new repository release and corresponding archive are needed to freeze the revised analysis publicly. The author should confirm the final release citation after depositing the supplied revision package.

Summary of remaining substantive limitations

No compatible regional observed-flow or field validation has been executed. Functional sovereignty is not measured. The graph remains provisional because junction, access, and operational information are incomplete. The 100-km coverage threshold is assumed. Greenfield grid-resolution, extreme tree-rule, and current protected-area tests are now supplied. Forest integrity, tenure, and constructability are not validated. The revision makes these boundaries explicit and asks for evaluation as a geospatial model-sensitivity study; it does not assert that the manuscript now meets every requested validation condition.

Additional evidence supplied on 7 October 2026

New manuscript Section II.H specifies repeated stresses, candidate-grid alignment, the complete-grid population potential, WorldCover sampling, and the current protected-area query. Section III.G and Tables IX–XI report the corresponding results. The new CSV/JSON files include repeated_joint_stress, candidate_grid_environment, candidate_grid_protection_counts, and jica_observed_screenline_benchmark. Raw WDPCA data are withheld from redistribution under producer terms. Their producer query and release are documented for lawful reacquisition.

The strongest outstanding objection remains empirical validation of the study’s regional accessibility or coverage target. Obtaining an observed traffic table does not validate a model that predicts a different quantity. We therefore do not claim all reviewer requests are satisfied, and respectfully seek evaluation as an exploratory geospatial sensitivity study. If compatible regional validation remains mandatory, that evidence must still be collected or obtained before the stronger claim can be supported.